TEMPORAL FACIAL ANALYSIS | ORIGINAL RESEARCH

FacialPulse: An Efficient RNN-based Depression Detection via Temporal Facial Landmarks

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CLINICAL / TECHNICAL MOTIVATION & OBJECTIVE

- Challenges: prior methods overlook the temporal dynamics of facial expressions, rely on computationally heavy 3D-CNN backbones, and use redundant raw-image features.
- Objective: build an efficient framework that models temporal facial-landmark motion with a bidirectional RNN, and calibrates landmark jitter, to detect depression fast and accurately.

SYSTEM DESIGN

Proposed Methodology: Two-Module Pipeline

STAGE 1 — FACIAL LANDMARK CALIBRATION MODULE (FLCM)

1 Landmark Extraction

Dlib / OpenFace detect 68 facial landmarks per frame after face detection & bounding-box crop.

2 Motion Prediction

Pyramid Lucas-Kanade optical flow predicts each landmark's next-frame position from prior motion.

3 Two-Way Error Denoise

Front-to-back vs. back-to-front flow is compared; landmarks with large discrepancy are flagged.

4 Kalman Filtering

Fuses the predicted position with the newly detected one, removing jitter, prediction & detection error.

STAGE 2 — FACIAL MOTION MODELING MODULE (FMMM)

Dual-Stream BiGRU

- Stream r1: absolute landmark positions per frame (sequence A) — captures overall facial geometry.
- Stream r2: frame-to-frame landmark differences (sequence B) — captures motion speed/dynamics.
- Both use Bidirectional GRU (k=64 units) to read past + future context; outputs merged via FC layers.

Why BiGRU?

- Depressed individuals show subtler, smoother facial motion — long-range temporal context is critical.
- Gate mechanism + parameter sharing mitigate vanishing gradients and enable parallel computation.
- Landmarks (not raw pixels) remove background/lighting redundancy, shrinking the model to 0.5M params.

CLINICAL PIPELINE

Datasets & Experimental Setup

AVEC2014 BENCHMARK [2]

- 300 webcam videos: "Northwind" (read German fable) + "FreeForm" (Q&A + childhood memory), 150 each.
- Merged & split: 240 training / 60 testing samples; ~2 min average duration.
- Labels: BDI-II questionnaire score, 0–63 (0–13 none, 14–19 mild, 20–28 moderate, 29–63 severe).
- Metrics: RMSE and MAE against ground-truth BDI-II scores.

MMDA (INTERNAL DATASET) [3]

- Multimodal Dataset for Depression and Anxiety — visual, acoustic & textual modalities from interviews.
- 300 depression-related video segments used (facial-expression modality only in this study).
- Diagnosis basis: combined Hamilton Rating Scale for Depression + Anxiety scores by professional psychologists.

IMPLEMENTATION DETAILS

- Landmark detectors: Dlib (primary), OpenFace (ablation) · BiGRU hidden units k = 64 · Dropout 0.25 (input) / 0.5 (remaining layers)
- Adam optimizer, learning rate 0.001 · Smooth L1 loss (outlier-robust) · 500 training epochs · Single RTX 3090 GPU, 24GB memory

BENCHMARKS

Experimental Evaluation & Benchmark Comparison

5.92
Best MAE on AVEC2014 (vs. 6.01 prior SOTA [4])

0.5M
Trainable parameters (vs. 24M for CNN+RNN)

2×
Faster training + preprocessing vs. baseline methods

11%
Reduction in landmark detection error after FLCM

AVEC2014 — RMSE / MAE VS. LEADING METHODS

Method	RMSE	MAE
Baseline — LGBP-TOP, SVR [2]	10.86	8.86
Melo et al. — MDN-152	7.65	6.06
Niu et al. — CNN+GCE+MSV [4]	7.56	6.01
FacialPulse (proposed)	**7.60**	**5.92**

ABLATION & EFFICIENCY

- FLCM ablation: removing calibration raises RMSE 7.60→8.34 and MAE 5.92→6.69; removing only the Kalman filter still costs accuracy (7.69 / 5.98).
- FMMM branches: combined (r1+r2) MAE 5.92 beats absolute-position-only (6.01) or motion-only (6.15) branches alone.
- Training cost: 0.5h vs. 9.5h (CNN) / 12.5h (CNN+RNN); GPU footprint 2.5GB vs. 6–9GB.
- On MMDA, FacialPulse reports MAE 4.35 vs. an SVM baseline's 3.97 [3].

FacialPulse · Best MAE among listed SOTA methods on AVEC2014 [2][4]; large training-time & parameter reduction

INSIGHTS

Major Findings & Clinical Takeaways

Temporal modeling drives accuracy

Facial expressions of depressed individuals change more smoothly over time; a bidirectional GRU that reads both past and future context captures this signature better than static, frame-based CNNs.

Calibration matters as much as modeling

Removing the Facial Landmark Calibration Module degrades RMSE/MAE the most of any ablation, confirming that jitter and detection noise materially corrupt temporal facial features.

Landmarks beat raw pixels for efficiency

Replacing raw images with 68-point landmarks cuts parameters to 0.5M and GPU use to 2.5GB, enabling 2× faster training without sacrificing accuracy.

Two complementary motion views

Modeling both absolute landmark position and frame-to-frame change (two BiGRU streams) outperforms either view alone, showing depression cues live in both geometry and dynamics.

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